COURSEWORK ASSIGNMENT SUBMISSION

Name: Daksh Srivastava **Roll No:** 150096725087
Subject: Software Engineering & Project Management **Day:** Day 10 — Functional and Non-Functional Requirements

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Section 11: Classroom Activity — Classify, Improve, and Test

Analysis of ambiguous and mixed requirements statements, classification, refinement into testable specifications, and verification design.

Original Raw Statement	Classification	Refined Measurable Requirement	Verification / Test Method
The system shall let students cancel a registration before the published deadline.	Functional Requirement	A registered student shall be able to cancel their registration up to exactly 24 hours prior to the event start timestamp via the portal dashboard.	Boundary Test: Submit cancellation request at T-24h 01m (accepted) and T-23h 59m (rejected with error).
The website must be very fast.	Ambiguous (Non-Functional)	Under a baseline network round-trip time of 50ms and 200 concurrent users, the home catalog page shall achieve a Time-to-First-Byte (TTFB) < 300ms and Largest Contentful Paint (LCP) < 1.8 seconds.	Automated Lighthouse CI audit and k6 synthetic load test scripts.
Only administrators may assign coordinator roles.	Constraint / Non-Functional (Security)	Access to role assignment endpoints shall require an authenticated user session possessing the `ROLE_SUPERADMIN` claim; unauthorized attempts must return HTTP 403 Forbidden.	Security Role Test: Attempt endpoint request using standard student token; assert HTTP 403 and security audit log entry.
The system shall support 300 concurrent users.	Non-Functional (Scalability)	The application server cluster shall sustain 300 simultaneous active HTTP user sessions executing typical browse-and-register actions without exceeding 1% CPU throttling or 0.1% HTTP 5xx error rate.	JMeter stress test ramping from 50 to 350 virtual users over a 15-minute window.
The system must use the college-approved sign-in	Constraint (Technical)	User identity authentication must	Architecture Inspection & Network trace asserting

service.		delegate strictly to the college OpenID Connect (OIDC) single sign-on provider; local database username/password storage is prohibited.	redirection to `sso.college.edu/oauth/authorize`.

Section 14: Homework — Requirements Catalogue & Acceptance Tests

Formal requirements specification and acceptance test suite for the College Event Management System (CEMS).

1. System Requirements Catalogue

Req ID	Category	Priority	Requirement Statement
FR-01	Functional	Must Have	The system shall allow users to search and filter events by title, date, venue, and host department.
FR-02	Functional	Must Have	The system shall enforce atomic seat allocation when a student registers, decrementing available seats by 1.
FR-03	Functional	Must Have	The system shall prevent a student from registering for two events whose start and end times overlap.
FR-04	Functional	Should Have	The system shall send an automated email confirmation containing a unique QR verification code upon successful registration.
FR-05	Functional	Should Have	The system shall allow event coordinators to manually check-in attendees via camera QR scanning or roll number entry.
FR-06	Functional	Could Have	The system shall allow students to download a digitally signed Certificate of Participation after event completion.
NFR-01	Performance	Must Have	The API shall respond to

			event registration requests within 1.2 seconds for the 95th percentile under 250 concurrent users.
NFR-02	Security	Must Have	All network transmissions must be encrypted using TLS 1.3; sensitive session tokens must carry HttpOnly and Secure cookie flags.
NFR-03	Reliability	Must Have	The database shall execute automated daily hot backups with a Recovery Point Objective (RPO) of less than 1 hour.
NFR-04	Usability	Should Have	The portal shall comply with WCAG 2.1 Level AA accessibility standards, supporting full keyboard navigation.
NFR-05	Compatibility	Must Have	The portal frontend shall render identically across modern versions of Chrome, Firefox, Safari, and Edge on mobile and desktop.

2. Detailed Acceptance Test Cases

- **Acceptance Test 1 (FR-02 Seat Allocation):** Precondition: Event capacity = 50, Registered count = 49. Action: Student A submits registration. Expected Result: Registration succeeds, screen displays confirmation token, database registered count becomes 50, and event card updates status badge to "Full".
- **Acceptance Test 2 (FR-03 Schedule Overlap):** Precondition: Student is registered for Event X (Friday 2:00 PM – 4:00 PM). Action: Student attempts to register for Event Y (Friday 3:00 PM – 5:00 PM). Expected Result: System blocks registration with message: "Schedule Conflict: You are already registered for Event X during this time."
- **Acceptance Test 3 (NFR-01 Performance Latency):** Precondition: Test harness configured with 250 simulated user threads. Action: Execute continuous catalog search and registration API hits for 10 minutes. Expected Result: 95% of response latency samples measure <= 1.200 seconds; zero 500 Internal Server errors logged.
- **Acceptance Test 4 (Security RBAC):** Precondition: User authenticated with standard student credentials. Action: Issue HTTP DELETE request to `/api/v1/events/102`. Expected Result: Request rejected with HTTP 403 Forbidden; system event audit log records security access violation.